

15.1.1

$$\begin{aligned} S_1 \cdot S_2 &= S_{1x} S_{2x} + S_{1y} S_{2y} + S_{1z} S_{2z} \\ &= \frac{(S_{1+} + S_{1-})}{2} \cdot \frac{(S_{2+} + S_{2-})}{2} + \frac{(S_{1+} - S_{1-})}{2i} \cdot \frac{(S_{2+} - S_{2-})}{2i} \\ &\quad + S_{1z} S_{2z} \end{aligned}$$

expanding, $= \frac{1}{2} (S_{1-} S_{2+} + S_{1+} S_{2-}) + S_{1z} S_{2z}$ (proving 15.1.12)

Then $S^2 = S_1^2 + S_2^2 + 2S_{1z} S_{2z} + S_{1-} S_{2+} + S_{1+} S_{2-}$

$$\begin{aligned} \rightarrow S^2 |++\rangle &= \hbar^2 \left(\frac{1}{2} \left(1 + \frac{1}{2}\right) + \frac{1}{2} \left(1 + \frac{1}{2}\right) + 2 \cdot \frac{1}{2} \cdot \frac{1}{2} + 0 + 0 \right) |++\rangle \\ &= 2\hbar^2 |++\rangle \end{aligned}$$

$$\begin{aligned} S^2 |+-\rangle &= \hbar^2 \left(\frac{1}{2} \left(1 + \frac{1}{2}\right) + \frac{1}{2} \left(1 + \frac{1}{2}\right) + 2 \left(\frac{1}{2}\right) \left(-\frac{1}{2}\right) \right) |+-\rangle \\ &\quad + \hbar^2 \left(-\sqrt{\left(\frac{1}{2} + \frac{1}{2}\right) \left(\frac{1}{2} - \frac{1}{2} + 1\right)} \right) \left(\sqrt{\left(\frac{1}{2} - \frac{1}{2}\right) \left(\frac{1}{2} - \frac{1}{2} + 1\right)} \right) |-+\rangle \\ &= \hbar^2 |+-\rangle + \hbar^2 |-+\rangle \end{aligned}$$

$$\begin{aligned} S^2 |-+\rangle &= \hbar^2 |-+\rangle + \hbar^2 \left(\sqrt{\left(\frac{1}{2} - \frac{1}{2}\right) \left(\frac{1}{2} - \frac{1}{2} + 1\right)} \right) \left(\sqrt{\left(\frac{1}{2} + \frac{1}{2}\right) \left(\frac{1}{2} - \frac{1}{2} + 1\right)} \right) |+-\rangle \\ &= \hbar^2 |-+\rangle + \hbar^2 |+-\rangle \end{aligned}$$

$$\begin{aligned} S^2 |--\rangle &= \hbar^2 \left(\frac{1}{2} \left(\frac{1}{2} + 1\right) + \frac{1}{2} \left(\frac{1}{2} + 1\right) + 2 \left(-\frac{1}{2}\right) \left(-\frac{1}{2}\right) \right) |--\rangle \\ &= 2\hbar^2 |--\rangle \end{aligned}$$

Since the $|++\rangle$, $|+-\rangle$, $|-+\rangle$, and $|--\rangle$ are orthogonal, we can infer the matrix form of S^2 in this basis:

$$S^2 = \hbar^2 \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \quad \text{(proving 15.1.10)}$$

(++) (+-) (-+) (--)

①

recall:
 $J_{\pm} |j, m\rangle$
 $= \sqrt{(j \mp m)(j \pm m + 1)} \hbar |j, m \pm 1\rangle$

finally, for 15.1.11, we write in matrix form

$$\frac{|+-\rangle + |-+\rangle}{\sqrt{2}} \rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\frac{|+-\rangle - |-+\rangle}{\sqrt{2}} \rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ \pm 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} \times 2 \text{ and } \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} \times (0)$$

→ eigenvectors of S^2 of eigenval $\hbar^2 s(s+1)$
where $s=1$ and 0 .

15.1.2 (1)

The total angular momentum basis diagonalizes H , because $H = H_{\text{coulomb}} + H_{\text{hf}} = H_{\text{coulomb}} + \frac{1}{2}A(S^2 - S_1^2 - S_2^2)$ and H_{coulomb} doesn't care about the spin of the state. Our ground states of H_{coulomb} , including spin labels, are $|n=1, l=0, m=0\rangle \otimes |s, s_z\rangle$ and we can have any of the 4 spin states because the proton and electron are distinguishable. The complete H eigenvals become

$$H |n=1\rangle \otimes |s, s_z\rangle = -R_y |n=1\rangle \otimes |s, s_z\rangle + H_{\text{hf}} |n=1\rangle \otimes |s, s_z\rangle \\ = \left[-R_y + \frac{1}{2}A \left(s(s+1) - \frac{1}{2}(\frac{1}{2}+1) - \frac{1}{2}(\frac{1}{2}+1) \right) \hbar^2 \right]$$

$\times |n=1\rangle \otimes |s, s_z\rangle$

$$\rightarrow E_{\pm} = \begin{cases} -R_y + \frac{\hbar^2 A}{4}, & s=1 \\ -R_y - \frac{3\hbar^2 A}{4}, & s=0 \end{cases}$$

②

$$15.2.2 \quad J_- |JM\rangle = \sqrt{J(J+1) - M(M-1)} |J, M-1\rangle$$

$\frac{1}{2} \otimes \frac{1}{2}$ $|\frac{3}{2}, \frac{3}{2}\rangle$ is the 1st $|JM\rangle$ state to consider,
because it can only be composed of 1 unique
 $|m_1, m_2\rangle$ state: $|m_1=j_1=\frac{1}{2}, m_2=j_2=\frac{1}{2}\rangle$

$$\rightarrow C_{JM, m_1 m_2}^{j_1 j_2} = \boxed{C_{\frac{3}{2}, \frac{3}{2}, \frac{1}{2} \frac{1}{2}}^{\frac{1}{2} \frac{1}{2}} = 1}$$

acting on $|JM\rangle$ with J_- and equivalently on $|m_1, m_2\rangle$ with $j_{1-} + j_{2-}$,

$$J_- \underbrace{|\frac{3}{2}, \frac{3}{2}\rangle}_{JM} = (j_{1-} + j_{2-}) \underbrace{|\frac{1}{2}, \frac{1}{2}\rangle}_{m_1 m_2} \quad \left\{ \begin{array}{l} \text{From now on, LHS will} \\ \text{always be } |JM\rangle \text{ basis,} \\ \text{and RHS will be } |m_1, m_2\rangle \end{array} \right.$$

$$\rightarrow \sqrt{\frac{3}{2}(\frac{3}{2}+1) - \frac{3}{2}(\frac{3}{2}-1)} |\frac{3}{2}, \frac{1}{2}\rangle = \sqrt{\frac{1}{2}(\frac{1}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} |-\frac{1}{2}, 1\rangle + \sqrt{1(1+\frac{1}{2}) - 1(1-1)} |\frac{1}{2}, 0\rangle$$

$$\rightarrow |\frac{3}{2}, \frac{1}{2}\rangle = \frac{1}{\sqrt{3}} |-\frac{1}{2}, 1\rangle + \sqrt{\frac{2}{3}} |\frac{1}{2}, 0\rangle$$

$$\Rightarrow \boxed{C_{\frac{3}{2}, \frac{1}{2}, -\frac{1}{2} 1}^{\frac{1}{2} \frac{1}{2}} = \frac{1}{\sqrt{3}}, \quad C_{\frac{3}{2}, \frac{1}{2}, \frac{1}{2} 0}^{\frac{1}{2} \frac{1}{2}} = \sqrt{\frac{2}{3}}}$$

$$\text{lowering again, } J_- |\frac{3}{2}, \frac{1}{2}\rangle = \frac{1}{\sqrt{3}} (j_{1-} |-\frac{1}{2}, 1\rangle + j_{2-} |-\frac{1}{2}, 1\rangle + \sqrt{\frac{2}{3}} j_{2-} |\frac{1}{2}, 0\rangle + \sqrt{2} j_{1-} |\frac{1}{2}, 0\rangle)$$

$$= \frac{1}{\sqrt{3}} \left(\sqrt{1(1+1) - 1(1-1)} |-\frac{1}{2}, 0\rangle + \sqrt{2} \sqrt{1(1+1) - 0(0-1)} |\frac{1}{2}, -1\rangle \right. \\ \left. + \sqrt{2} \sqrt{\frac{1}{2}(\frac{1}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} |-\frac{1}{2}, 0\rangle \right)$$

$$= \sqrt{\frac{2}{3}} |-\frac{1}{2}, 0\rangle + \sqrt{\frac{2}{3}} |-\frac{1}{2}, 0\rangle + \sqrt{\frac{2}{3}} \sqrt{2} |\frac{1}{2}, -1\rangle$$

$$= \sqrt{\frac{2}{2}(\frac{3}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} |J=\frac{3}{2}, M=-\frac{1}{2}\rangle$$

$$\rightarrow \boxed{C_{\frac{3}{2}, -\frac{1}{2}, -\frac{1}{2} 0}^{\frac{1}{2} \frac{1}{2}} = \sqrt{\frac{2}{3}}}$$

$$C_{\frac{3}{2}, -\frac{1}{2}, \frac{1}{2} -1}^{\frac{1}{2} \frac{1}{2}} = \frac{1}{\sqrt{3}}$$

(3)

Finally, $J_- \left| \frac{3}{2}, -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} (j_- + j_-) \left| -\frac{1}{2}, 0 \right\rangle + \frac{1}{\sqrt{3}} (j_- + j_-) \left| \frac{1}{2}, -1 \right\rangle$

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$$\sqrt{\frac{3}{2}(\frac{3}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} \left| \frac{3}{2}, -\frac{3}{2} \right\rangle = \sqrt{\frac{2}{3}} \sqrt{1(1+1) - 0(0-1)} \left| -\frac{1}{2}, -1 \right\rangle + \frac{1}{\sqrt{3}} \sqrt{\frac{1}{2}(\frac{1}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} \left| \frac{1}{2}, -1 \right\rangle$$

$$\rightarrow \left| \frac{3}{2}, -\frac{3}{2} \right\rangle = \frac{1}{\sqrt{3}} \left(\sqrt{\frac{2}{3}} \sqrt{2} \left| -\frac{1}{2}, -1 \right\rangle + \frac{1}{\sqrt{3}} \left| \frac{1}{2}, -1 \right\rangle \right)$$

$$= \frac{1}{\sqrt{3}} \left| -\frac{1}{2}, -1 \right\rangle \rightarrow \boxed{C_{\frac{3}{2}, -\frac{3}{2}, -\frac{1}{2}, -1}^{\frac{1}{2}, 1} = 1}$$

This takes care of $J = \frac{3}{2}$, but for $J = \frac{1}{2}$ we must determine the starting state $\left| \frac{1}{2} = J, M = \frac{1}{2} \right\rangle = A \left| m_1 = \frac{1}{2}, m_2 = 0 \right\rangle + B \left| m_1 = -\frac{1}{2}, m_2 = 1 \right\rangle$ by demanding (1) Normalization: $A^2 + B^2 = 1$, and (2) orthogonality with the $M = \frac{1}{2}$ state in the $J = \frac{3}{2}$ 4-plet.

$$(2) \rightarrow \frac{B}{\sqrt{3}} + \sqrt{\frac{2}{3}} A = \left\langle \frac{3}{2}, \frac{1}{2} \left| \frac{1}{2}, \frac{1}{2} \right\rangle = 0$$

$$(1) \rightarrow A = \pm \sqrt{1 - B^2}$$

$$\Rightarrow -\frac{B}{\sqrt{3}} = \sqrt{\frac{2}{3}} \sqrt{1 - B^2} \Rightarrow \frac{B^2}{3} = \frac{2}{3} (1 - B^2)$$

$$\Rightarrow B^2 = \frac{2}{3}, A^2 = \frac{1}{3}$$

for orthogonality, then, take $B = \sqrt{\frac{2}{3}}$ and the negative root $A = -\frac{1}{\sqrt{3}}$ (there is an overall irrelevant phase choice)

$$\rightarrow \left| \frac{1}{2}, \frac{1}{2} \right\rangle = -\frac{1}{\sqrt{3}} \left| \frac{1}{2}, 0 \right\rangle + \sqrt{\frac{2}{3}} \left| -\frac{1}{2}, 1 \right\rangle \Rightarrow \boxed{C_{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0}^{\frac{1}{2}, 1} = -\frac{1}{\sqrt{3}}; C_{\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, 1}^{\frac{1}{2}, 1} = \sqrt{\frac{2}{3}}}$$

lowering, $J_- \left| \frac{1}{2}, \frac{1}{2} \right\rangle = (j_- + j_-) \left(-\frac{1}{\sqrt{3}} \left| \frac{1}{2}, 0 \right\rangle + \sqrt{\frac{2}{3}} \left| -\frac{1}{2}, 1 \right\rangle \right)$

$$\rightarrow \sqrt{\frac{1}{2}(\frac{1}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\frac{1}{\sqrt{3}} \sqrt{\frac{1}{2}(\frac{1}{2}+1) - \frac{1}{2}(\frac{1}{2}-1)} \left| -\frac{1}{2}, 0 \right\rangle - \frac{1}{\sqrt{3}} \sqrt{1(1+1) - 0(0-1)} \left| \frac{1}{2}, -1 \right\rangle$$

$$+ \sqrt{\frac{2}{3}} \sqrt{1(1+1) - 1(1-1)} \left| -\frac{1}{2}, 0 \right\rangle$$

(4)

$$\Rightarrow \left| \frac{1}{2} \frac{1}{2} \right\rangle = \frac{1}{\sqrt{3}} \left| \frac{1}{2} 0 \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2} -1 \right\rangle$$

$$\Rightarrow \boxed{C_{\frac{1}{2} \frac{1}{2} \frac{1}{2} 0}^{\frac{1}{2} 1} = -\sqrt{\frac{2}{3}}, \quad C_{\frac{1}{2} \frac{1}{2} \frac{1}{2} 0}^{\frac{1}{2} 1} = \frac{1}{\sqrt{3}}}$$

This completes the $J = \frac{1}{2}$ doublet.

101

Begin with $|J=2, M=2\rangle = |m_1=1, m_2=1\rangle$

$$\Rightarrow \boxed{C_{2211}'' = 1}$$

Lowering once,

$$|2, 1\rangle = |10\rangle \frac{1}{\sqrt{2}} + |01\rangle \frac{1}{\sqrt{2}}$$

$$\rightarrow \boxed{C_{2110}'' = C_{2101}'' = \frac{1}{\sqrt{2}}}$$

Lowering again,

$$|2, 0\rangle = \frac{1}{\sqrt{6}} |1-1\rangle + \sqrt{\frac{2}{3}} |00\rangle + \sqrt{\frac{1}{6}} |-11\rangle$$

$$\rightarrow \boxed{C_{201-1}'' = C_{20-11}'' = \sqrt{\frac{1}{6}}, \quad C_{2000}'' = \sqrt{\frac{2}{3}}}$$

and again,

$$|2, -1\rangle = \frac{1}{\sqrt{2}} |0-1\rangle + \frac{1}{\sqrt{2}} |-10\rangle$$

$$\rightarrow \boxed{C_{2-10-1}'' = C_{2-1-10}'' = \frac{1}{\sqrt{2}}}$$

$$\text{and finally, } |2, -2\rangle = |-1-1\rangle \Rightarrow \boxed{C_{2-2-1-1}'' = 1}$$

Now the $J=1$ triplet begins with $|J=1, M=1\rangle$ and as in part ① must be determined by normalization and orthogonality with $|J=2, M=1\rangle$

$$\Rightarrow |J=1, M=1\rangle = \frac{1}{\sqrt{2}} |m_1=1, m_2=0\rangle + \frac{-1}{\sqrt{2}} |m_1=0, m_2=1\rangle$$

$\uparrow$
 C''_{1110}

$\uparrow$
 C''_{1101}

$$\text{lowering, } |10\rangle = \frac{1}{\sqrt{2}} |1-1\rangle + 0|00\rangle - \frac{1}{\sqrt{2}} |-11\rangle$$

$\uparrow$
 C''_{101-1}

$\uparrow$
 C''_{1000}

$\uparrow$
 C''_{10-11}

$$\text{lowering, } |1-1\rangle = \frac{1}{\sqrt{2}} |0-1\rangle - \frac{1}{\sqrt{2}} |-10\rangle$$

$\uparrow$
 C''_{1-10-1}

$\uparrow$
 C''_{1-1-10}

Finally, the $J=0$ singlet is determined by orthog with the remaining $M=0$ states, $|J=2, M=0\rangle$ and $|J=1, M=0\rangle$ (and by its normalization.)

$$\rightarrow |J=0, M=0\rangle = \frac{1}{\sqrt{3}} |1-1\rangle - \frac{1}{\sqrt{3}} |00\rangle + \frac{1}{\sqrt{3}} |-11\rangle$$

$\uparrow$
 C''_{001-1}

$\uparrow$
 C''_{0000}

$\uparrow$
 C''_{00-11}

⑥

$$\boxed{15.2.6} \quad (L+S)^2 = J^2 \rightarrow L \cdot S = \frac{1}{2}(J^2 - L^2 - S^2) \\ = \frac{1}{2}(j(j+1) - l(l+1) - s(s+1))$$

$s = \frac{1}{2}$ so we have 2 subspaces: $j = l \pm \frac{1}{2}$

$$\rightarrow 2L \cdot S = \begin{cases} (l + \frac{1}{2})(l + \frac{3}{2}) - l(l+1) - \frac{3}{4} = l, & j = l + \frac{1}{2} \\ (l - \frac{1}{2})(l + \frac{1}{2}) - l(l+1) - \frac{3}{4} = -l-1, & j = l - \frac{1}{2} \end{cases}$$

Now an arbitrary state can be expanded:

$$|\psi\rangle = \sum_{\substack{[l-s] \leq j_z \leq l+s \\ -j \leq j_z \leq j}} C_{jj_z}^{\psi} |l, s, j, j_z\rangle \langle l, s, j, j_z | \psi \rangle$$

$$\rightarrow 2L \cdot S |\psi\rangle = \sum_{[-l-\frac{1}{2} \leq j_z \leq l+\frac{1}{2}]} l \cdot C_{l+\frac{1}{2}, j_z}^{\psi} |l, s, l+\frac{1}{2}, j_z\rangle \langle l, s, l+\frac{1}{2}, j_z | \psi \rangle$$

$$+ \sum_{[l+\frac{1}{2} \leq j_z \leq l-\frac{1}{2}]} (-l-1) C_{l-\frac{1}{2}, j_z}^{\psi} |l, s, l-\frac{1}{2}, j_z\rangle \langle l, s, l-\frac{1}{2}, j_z | \psi \rangle$$

$$\rightarrow (2L \cdot S - l\mathbb{I}) |\psi\rangle = \sum_{[l+\frac{1}{2} \leq j_z \leq l-\frac{1}{2}]} (l-1-l) C_{l-\frac{1}{2}, j_z}^{\psi} |l, s, l-\frac{1}{2}, j_z\rangle \langle l, s, l-\frac{1}{2}, j_z | \psi \rangle$$

$$\rightarrow \boxed{P_- = \frac{2L \cdot S - l}{-2l-1}}$$

$$\text{and } 2L \cdot S - (-l-1)\mathbb{I} |\psi\rangle = \sum_{[l-\frac{1}{2} \leq j_z \leq l+\frac{1}{2}]} (l-(-l-1)) C_{l+\frac{1}{2}, j_z}^{\psi} |l, s, l+\frac{1}{2}, j_z\rangle \langle l, s, l+\frac{1}{2}, j_z | \psi \rangle$$

$$\rightarrow \boxed{P_+ = \frac{2L \cdot S + l+1}{2l+1}}$$